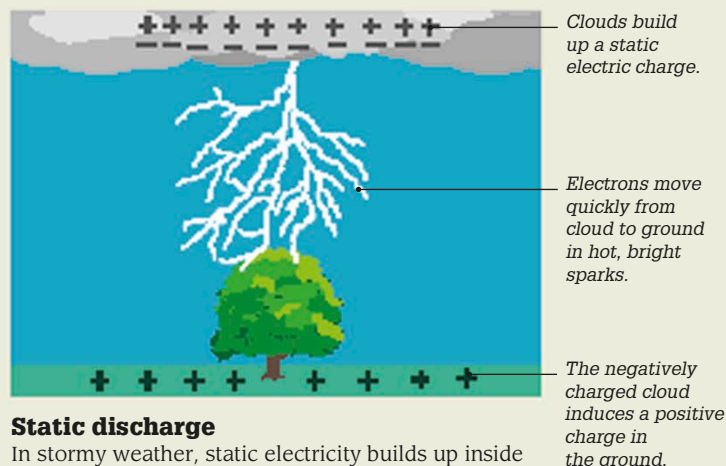


Electricity

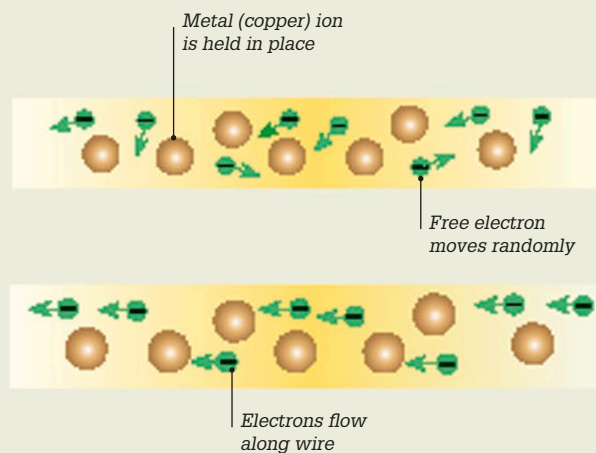
Electricity is a type of energy that is produced by the movement of electrons—tiny particles found in atoms. When the electrons flow through a material—such as a copper wire—it is known as current electricity. If the electricity builds up in one place, it is known as static



Static discharge

In stormy weather, static electricity builds up inside thunderclouds, and the base of the cloud becomes negatively charged. This induces a positive charge in the ground, and the attraction between the two creates a huge spark: a lightning bolt.

electricity. Electric current can only flow through materials called conductors. Many metals make good conductors as they contain free electrons that can move easily.

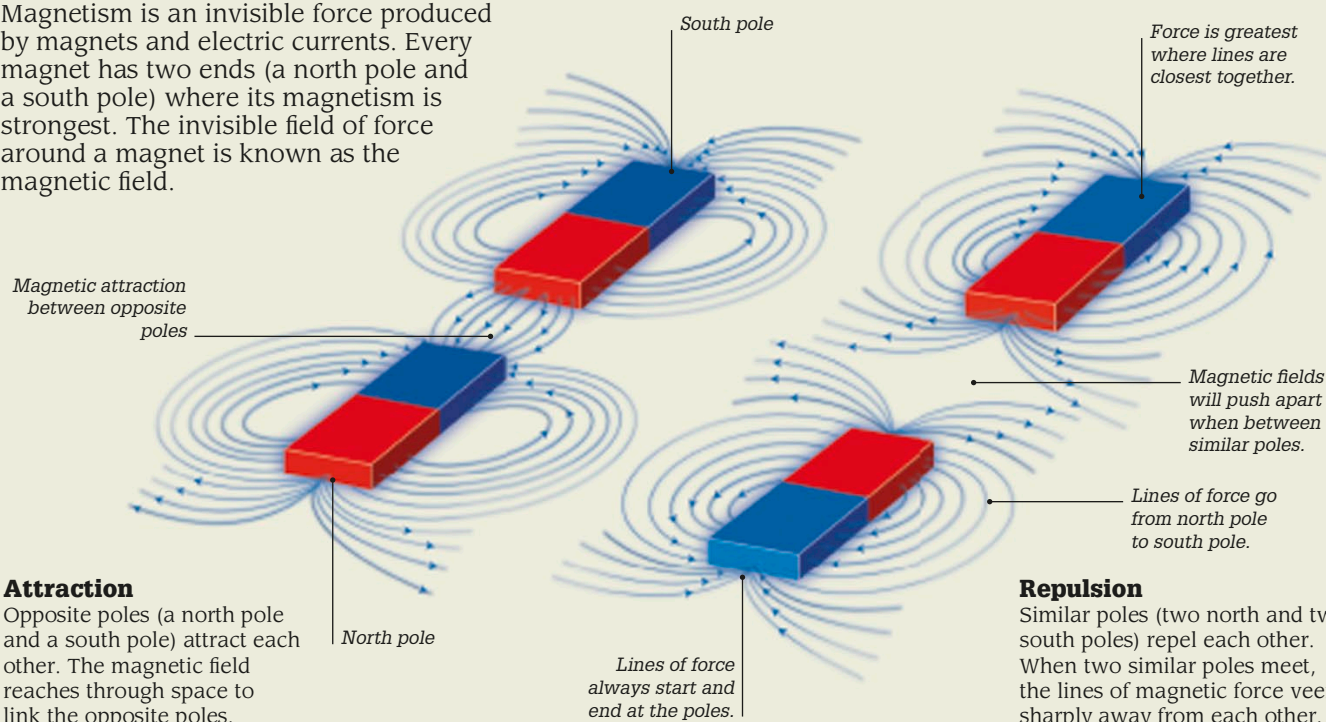


Electric currents

Current electricity flows around a path called a circuit. The electrons need a power source, such as a battery, to provide them with the energy to flow through the circuit. If there is no connection to a battery, the free electrons will move randomly in all directions.

Magnets

Magnetism is an invisible force produced by magnets and electric currents. Every magnet has two ends (a north pole and a south pole) where its magnetism is strongest. The invisible field of force around a magnet is known as the magnetic field.



Attraction

Opposite poles (a north pole and a south pole) attract each other. The magnetic field reaches through space to link the opposite poles.

Repulsion

Similar poles (two north and two south poles) repel each other. When two similar poles meet, the lines of magnetic force veer sharply away from each other.